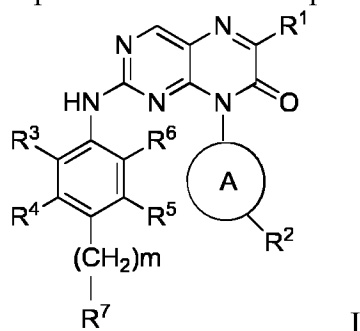


Claims

1. A compound of formula I or a pharmaceutical acceptable salt thereof:



5 wherein

R^1 is independently selected from a group consisting of a hydrogen, substituted or unsubstituted C_1 - C_{10} alkyl, C_2 - C_6 alkenyl, C_2 - C_6 alkynyl, optionally substituted C_3 - C_8 cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted benzyl, substituted or unsubstituted heterocyclyl, substituted or unsubstituted arylheterocyclyl;

10 A is a divalent radical $-(CR_aR_b)_n-A'$ - or absent, wherein R_a and R_b are independently selected from a group consisting of H, a C_{1-3} alkyl, a halogen, and n is an integer from 0 to 3;

A' is a benzene ring, a five- or six-membered heterocycle or a C_3 - C_8 cycloalkyl;

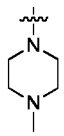
15 R^2 is independently selected from a group consisting of a hydrogen, unsubstituted or halogen-substituted C_1 - C_4 alkyl, nitro, amino, halogen, C_1 - C_6 alkoxy, optionally substituted acyloxy, optionally substituted acylamino, optionally substituted acyl; wherein, when A' is a benzene ring, R^2 is meta-substituted;

m is an integer from 0 to 3;

20 R^7 is independently selected from a substituted or unsubstituted NH_2 , a substituted or unsubstituted heterocyclyl, a substituted or unsubstituted arylheterocyclyl, a substituted or unsubstituted C_1 - C_{10} alkyl;

25 R^3 , R^4 , R^5 and R^6 are independently selected from a group consisting of H, a halogen, a substituted or unsubstituted C_1 - C_6 (preferably C_1 - C_3)alkoxy, a substituted or unsubstituted C_1 - C_6 (preferably C_1 - C_3)alkyl, NR_cR_d ; wherein each of R_c and R_d is independently selected from C_1 - C_3 alkyl;

wherein, when R^1 is H and m is 0, R^3 , R^4 , R^5 and R^6 are not H at the same time;

when R^1 is H, m is 0, A' is a benzene ring and R^7 is , if one of R^3 and R^6 is a methoxy, the other can not be H.

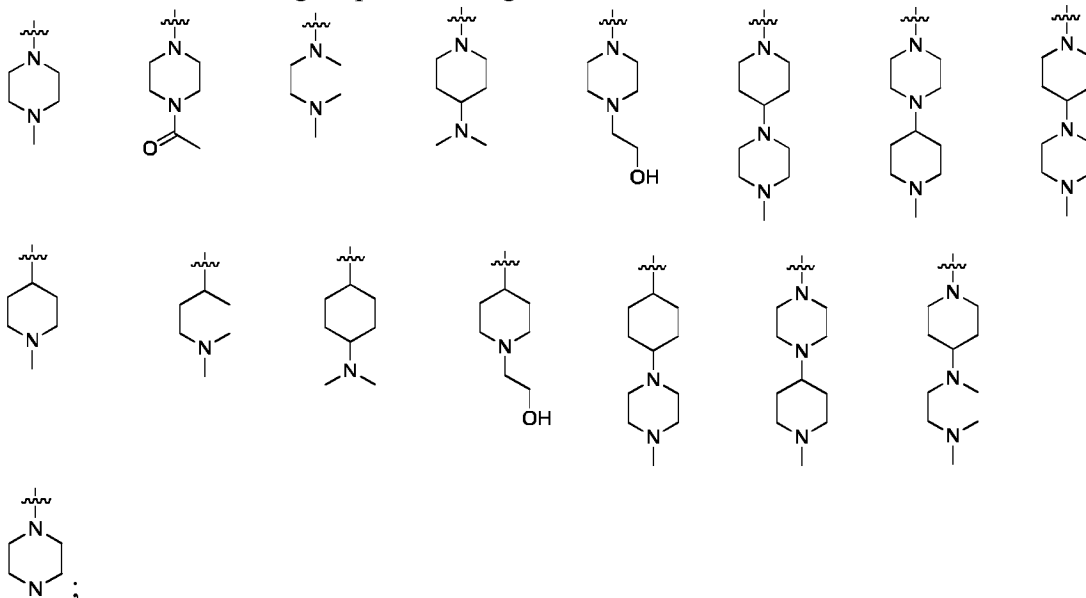
30 2. The compound of formula I or a pharmaceutical acceptable salt thereof of claim 1, wherein

R¹ is independently selected from a hydrogen, substituted or unsubstituted C₁-C₁₀ alkyl (preferably C₁-C₆ alkyl), substituted or unsubstituted phenyl, substituted or unsubstituted benzyl, optionally substituted C₃-C₈ cycloalkyl;

A' is a benzene ring, a five- or six-membered nitrogen-containing heterocycle;

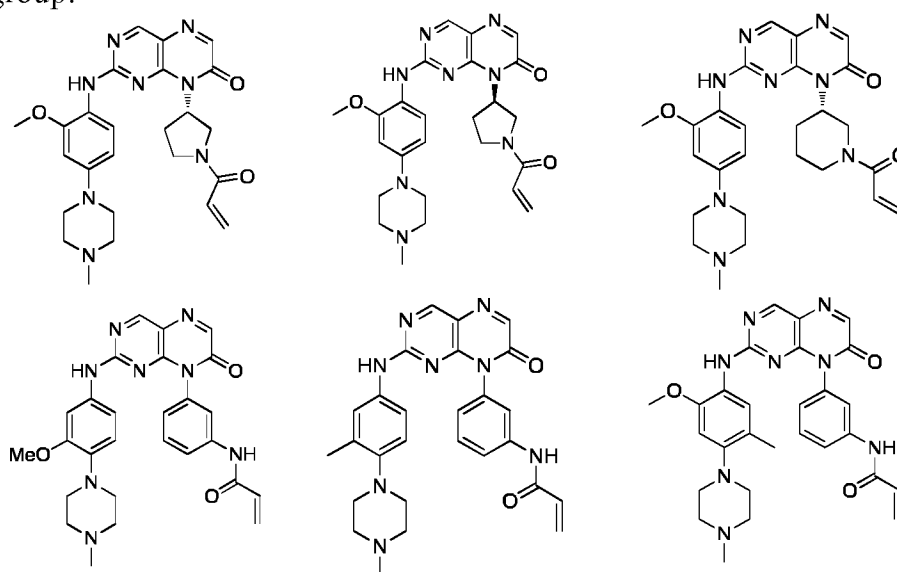
5 R² is independently selected from a C₁-C₃ alkyl, C₂-C₄ alkenyl substituted acylamino, substituted or unsubstituted C₂-C₄ alkenyl substituted acyl;

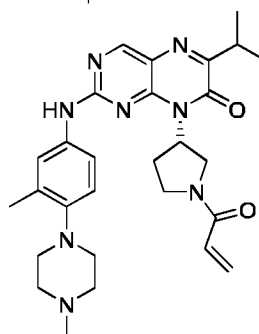
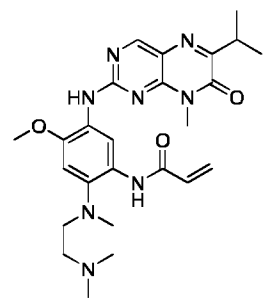
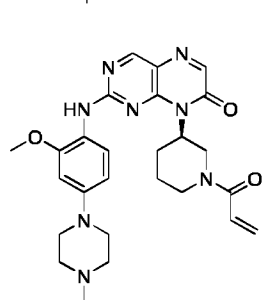
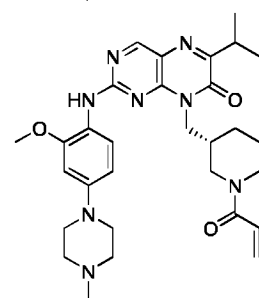
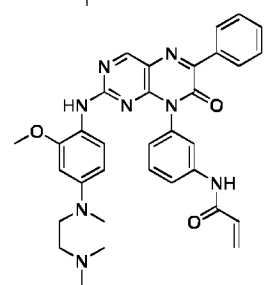
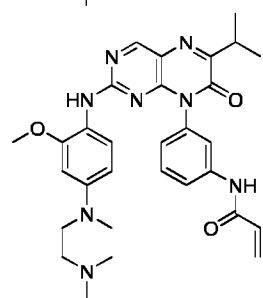
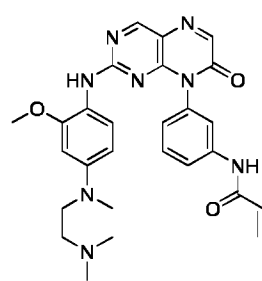
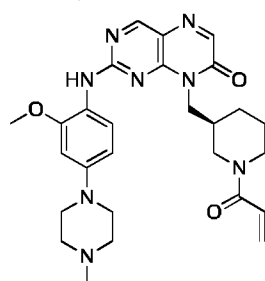
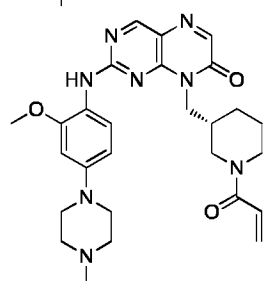
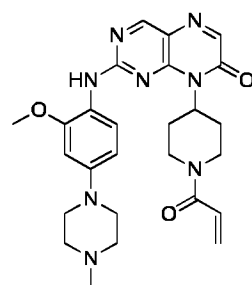
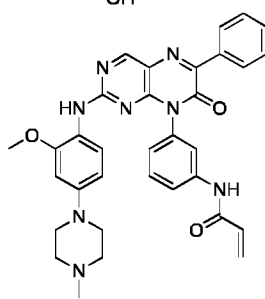
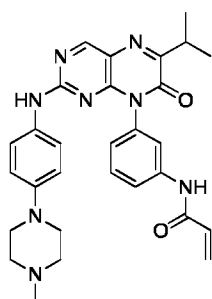
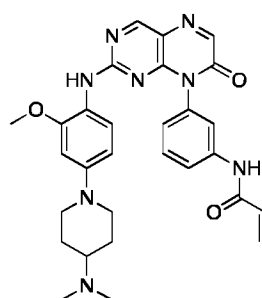
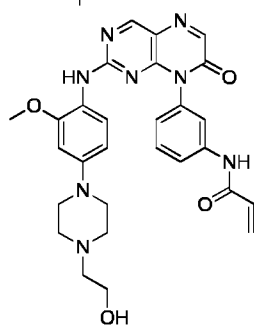
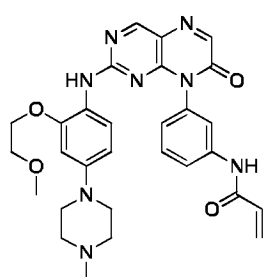
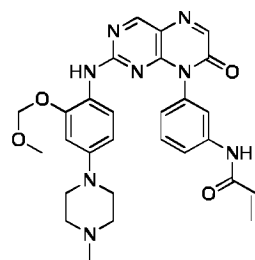
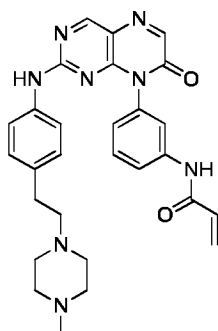
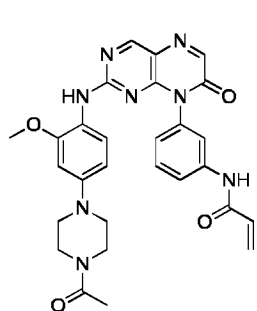
R⁷ is selected from a group consisting of:

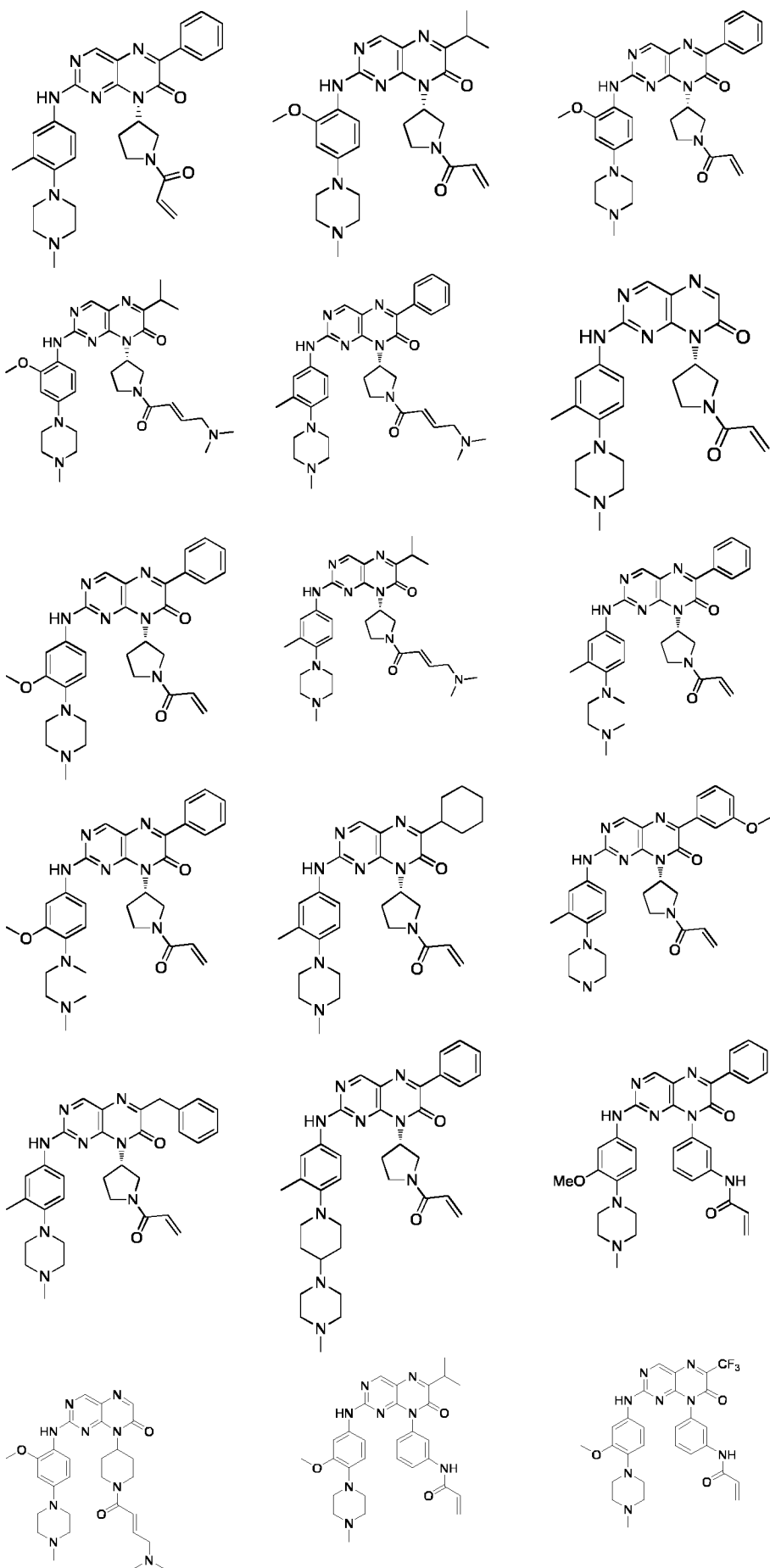


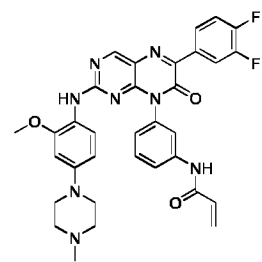
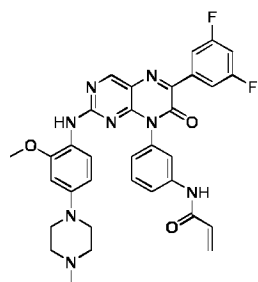
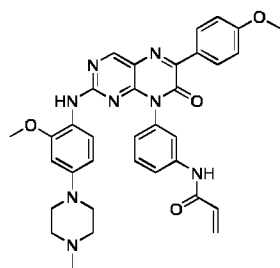
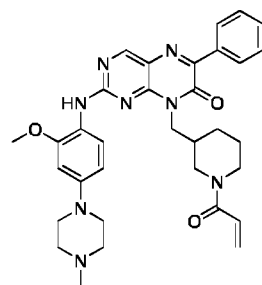
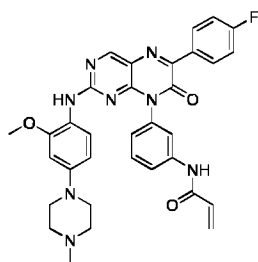
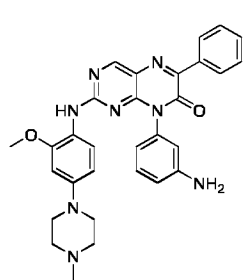
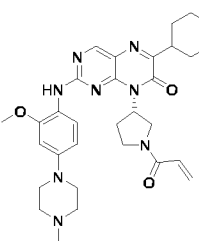
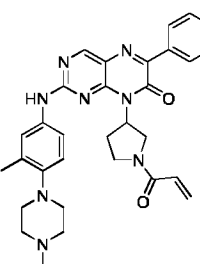
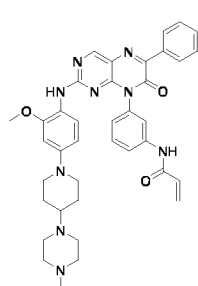
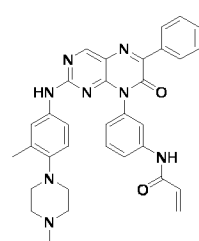
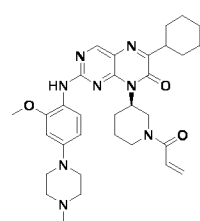
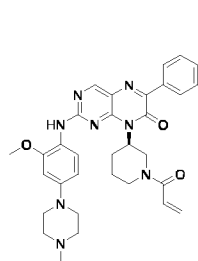
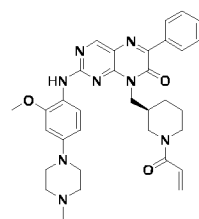
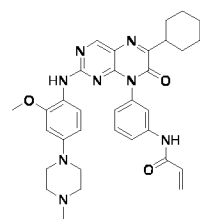
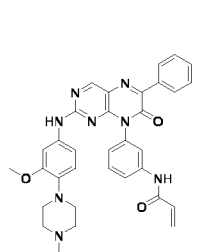
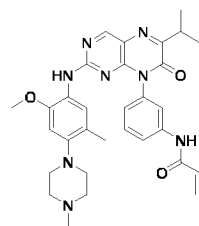
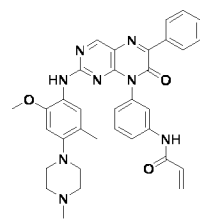
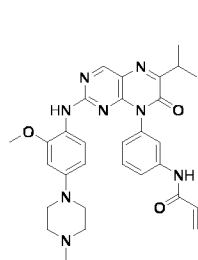
10 R³, R⁴, R⁵ and R⁶ are independently selected from a group consisting of H, a substituted or unsubstituted C₁-C₆ (preferably C₁-C₃)alkoxy, a substituted or unsubstituted C₁-C₆ (preferably C₁-C₃)alkyl.

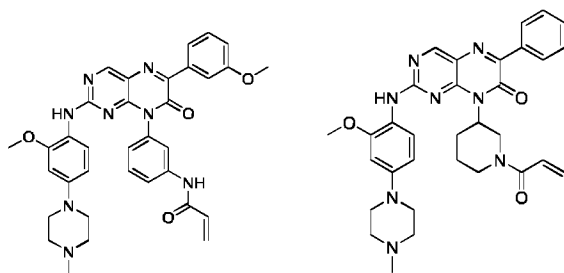
3. A compound or a pharmaceutical acceptable salt thereof selected from the following group:











4. The compound of claim 1, wherein

R^1 is independently selected from a group consisting of a hydrogen, substituted or unsubstituted C_1 - C_6 alkyl, or substituted or unsubstituted phenyl;

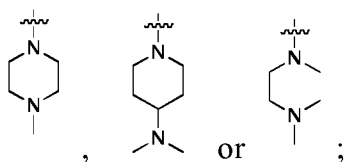
$n = 0$;

5 A' is a benzene ring, or a five-membered nitrogen-containing heterocycle;

R^2 is independently selected from a C_2 - C_4 alkenyl substituted acylamino, or C_2 - C_4 alkenyl substituted acyl;

m is an integer from 0 to 2;

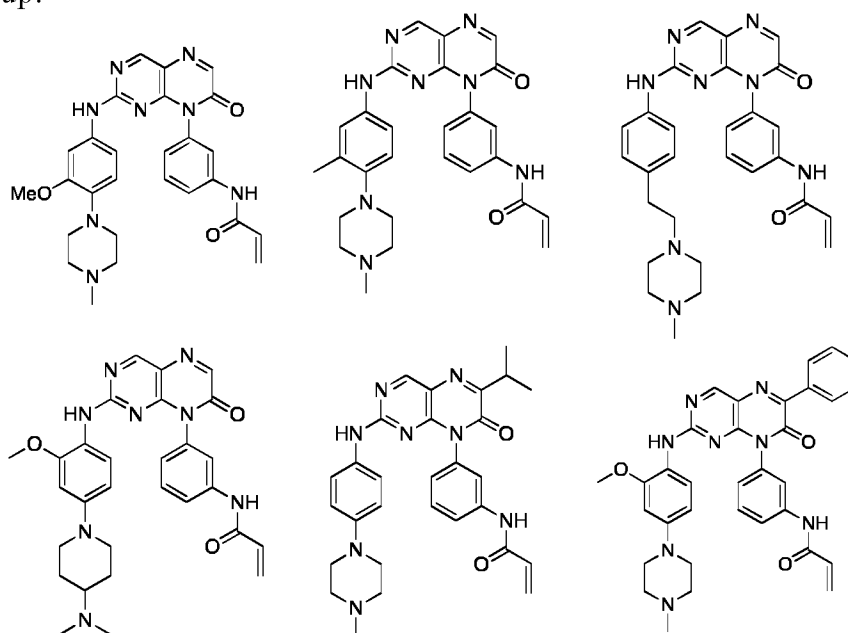
R^7 is selected from:

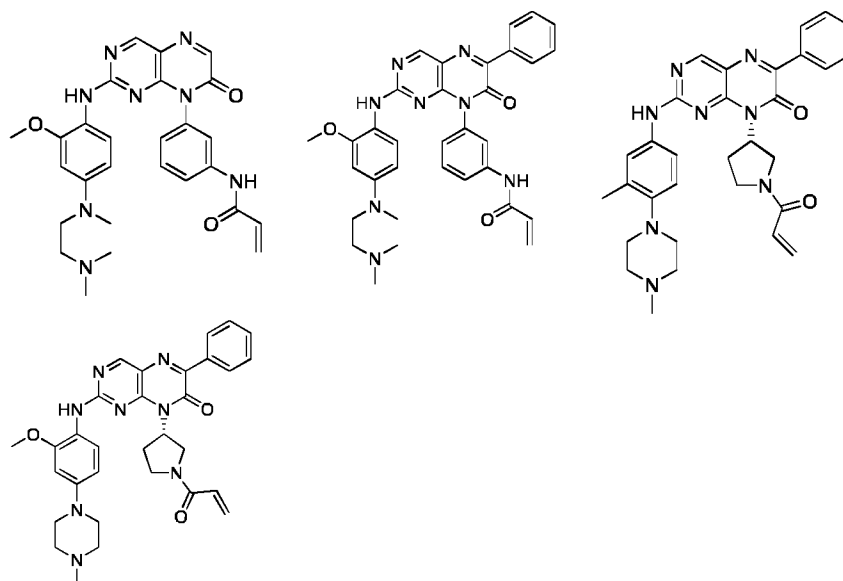


10

R^3 , R^4 , R^5 and R^6 are independently selected from a group consisting of H, a substituted or unsubstituted C_1 - C_3 alkoxy, a substituted or unsubstituted C_1 - C_3 alkyl.

5. A compound or a pharmaceutical acceptable salt thereof selected from the following group:





6. A pharmaceutical composition, comprising the compound or a pharmaceutical acceptable salt thereof of any one of claims 1-5 and a pharmaceutically acceptable carrier or excipient.

7. Use of the compound of any one of claims 1-5 for preparation of a medicament for treating or preventing EGFR-mediated diseases, or inhibiting EGFR.

8. The use of claim 7, wherein the EGFR-mediated disease is cancer.

9. The use of claim 8, wherein the cancer is selected from a group consisting of non-small cell lung cancer, small cell lung cancer, lung adenocarcinoma, lung squamous cell carcinoma, breast cancer, prostate cancer, glioblastoma, ovarian cancer, squamous cell carcinoma of head and neck, cervical cancer, esophageal cancer, liver cancer, kidney cancer, pancreas cancer, colon cancer, skin cancer, leukemia, lymphoma, stomach cancer, multiple myeloma and solid tumors.

10. A method for treating or preventing EGFR-mediated diseases by using the compound of any one of claims 1-5.